



Computer Network

File Transfer Protocol
(FTP)



FTP introduction

- The File Transfer Protocol (FTP) is an application layer protocol. FTP is an unusual service in that it utilizes two ports, a 'Data' port and a 'CMD' (command) port. Traditionally these are port 21 for the command port and port 20 for the data port.
- FTP can be used in two modes, active and passive. Depending on the mode, the data port is not always on port 20.

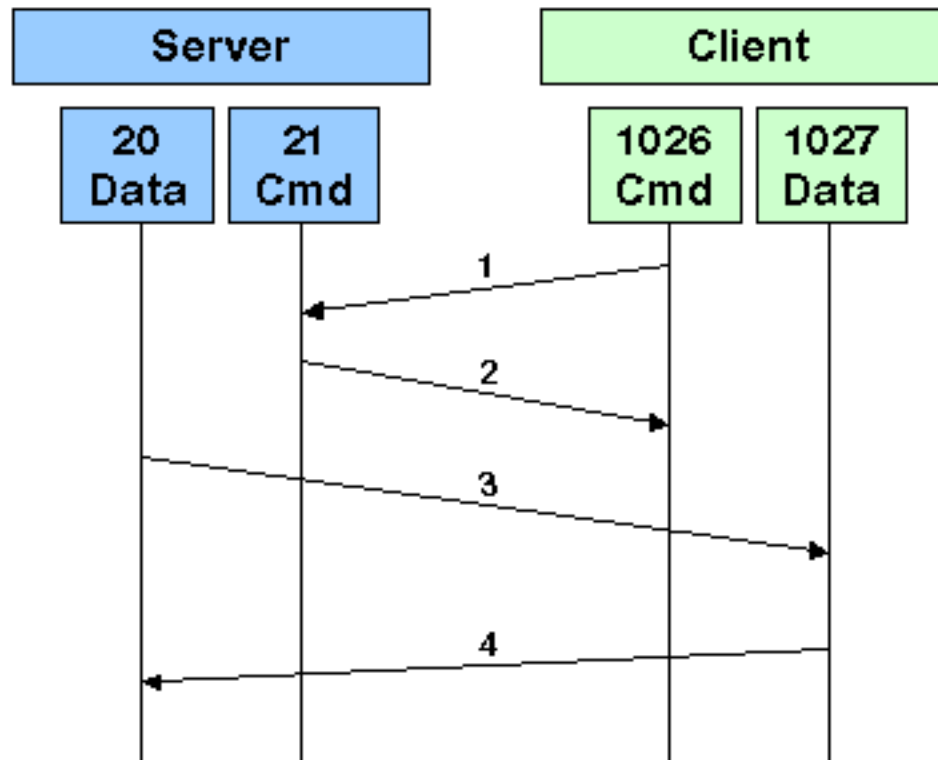


Active mode FTP

- In active mode FTP the client connects from a random unprivileged port P ($P > 1023$) to the FTP server's command port, port 21. Then, the client starts listening to port $P+1$ and sends the FTP command `PORT P+1` to the FTP server. The server will then connect back to the client's specified data port from its local data port, which is port 20.



Active mode FTP



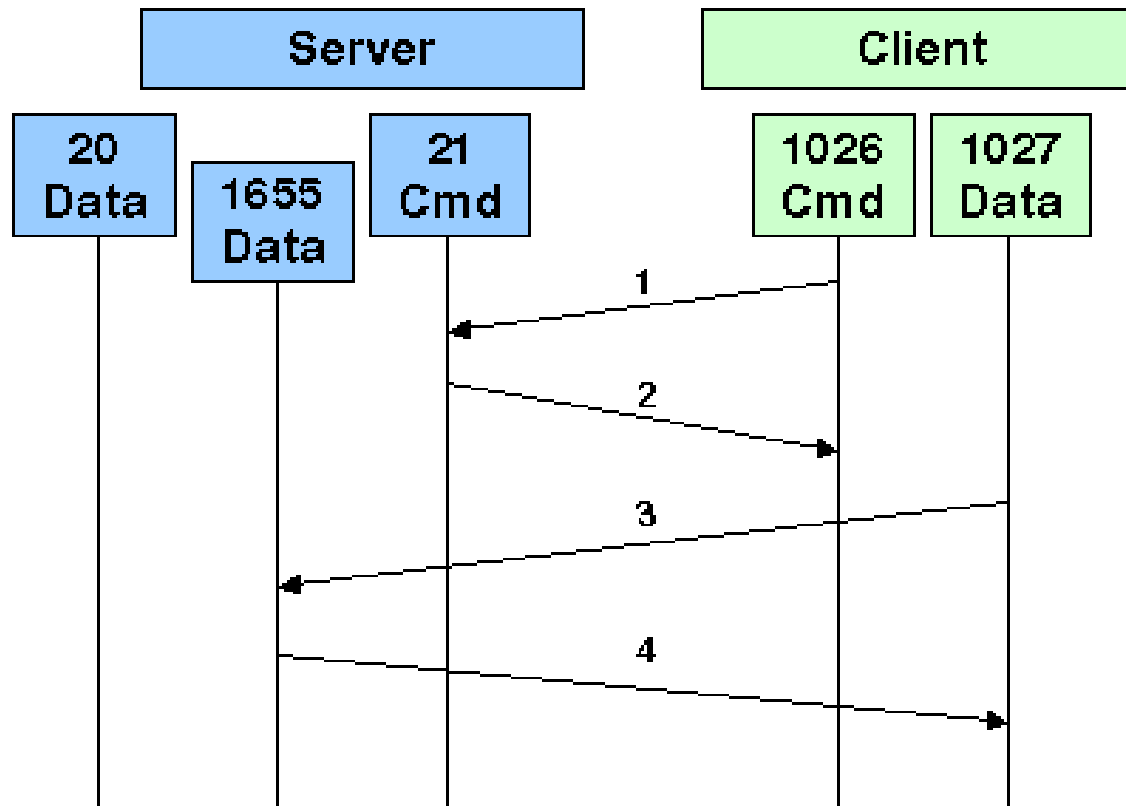


Passive mode FTP

- In passive mode FTP the client connects from a random unprivileged port P ($P > 1023$) to the FTP server's command port, port 21. The client sends in passive mode the PASV command to the server and expects an answer with the information on which port the server is listening for the data connection.
- After receiving this information, the client connects to the specified data port of the server from its local data port.



Passive mode FTP





FTP Service



- In Windows server 2008 (IIS)
- In Linux (VsFTP)
- In third party : Ser-U FTP, FileZilla